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Markov Chains

Continuously updated notes

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Chapter 1

Introduction

1.1 Introduction to Markov Chains

Definition 1.1.1. Let Ω be a sample space with a σ-algebra A defined on it. Then (Ω, A) is called a measurable space, and $\mu : A \to [0; \infty]$ is called a measure if $\mu(\emptyset) = 0$ and for disjoint $A_i \in A$,

$$\mu(\bigcup A_i) = \Sigma\mu(A_i),$$

and the ordered triple (Ω, A, μ) is called a measure space.

Definition 1.1.2. The smallest σ-algebra containing the open subsets of $\mathbb{R}$ is called the system of Borel-measurable sets, whose elements are the Borel-measurable sets (if the base set is not $\mathbb{R}$, then it is the smallest σ-algebra containing the open sets of the given topological space).

Definition 1.1.3. A function $f : \Omega \to \mathbb{R}$ defined on a measurable space (Ω, A) is a measurable function if the preimage under f of every Borel-measurable subset of $\mathbb{R}$ is measurable, i.e. belongs to A. If $P : A \to [0; 1]$ is a measure with $P(\Omega) = 1$, then the triple (Ω, A, P) is called a probability space.

Definition 1.1.4. A function $X : \Omega \to \mathbb{R}$ defined on the sample space Ω of a probability space (Ω, A, P) is a random variable (this is the official definition) if for every $x \in \mathbb{R}$,

$$\{\omega \in \Omega : X(\omega) \leq x\} \in A$$

Remark 1.1.5. Generally, by a real- or vector-valued random variable we mean a measurable function taking values in $(\mathbb{R}^n, \mathbb{B}^n)$, defined on a probability space (Ω, A, P), where $\mathbb{B}^n$ is the σ-algebra of Borel-measurable sets in $\mathbb{R}^n$. Unless stated otherwise, the Borel sets of a set I are denoted by $\mathbb{B}_I$ or $\mathbb{B}$ (the latter only when the notation is unambiguous).

Definition 1.1.6. The distribution function of a random variable X defined on the probability space (Ω, A, P) is $F_X : \mathbb{R} \to \mathbb{R}$, $F_X(x) = P(X < x)$. A function $f : \mathbb{R} \to \mathbb{R}$ is the density function of the absolutely continuous random variable X if $F_X(x) = \int_{-\infty}^{x} f(t)dt$.

Definition 1.1.7. A two-variable function

$$X(t, \omega) : T \times \Omega \to \mathbb{R}$$

defined on the probability space (Ω, A, P) and its time set T is a **stochastic process** if for every fixed $t \in T$, $X(t, \omega)$ is a random variable on (Ω, A, P). In the case $T = \mathbb{N}$, $X(t; \omega)$ is a discrete-time process, and in the case $T = [0; \infty)$ it is a continuous-time stochastic process. For a fixed $\omega_0 \in \Omega$,

$$X(t, \omega_0) : T \to \mathbb{R}$$

is called the **trajectory** of the stochastic process $X(t;\omega) : T \times \Omega \to \mathbb{R}$ under outcome ω_0.

Definition 1.1.8.

$$X(n;\omega) = X_n(\omega) : \mathbb{N} \times \Omega \to I$$

is a (discrete-time) **Markov chain / Markov process** if it has the **Markov property**, i.e., with

$$A_n = \sigma(X_0, X_1, ..., X_n)$$

for every $B \in \mathbb{B}_I$ and every $m \geq n$,

$$P(X_m \in B | X_0, X_1, ..., X_n) = P(X_m \in B | A_n) = P(X_m \in B | X_n)$$

Definition 1.1.9.

$$X_t(\omega) = X(t;\omega) : [0, \infty] \times \Omega \to I$$

is a continuous-time stochastic process, a (continuous-time) **Markov process**, if it has the **Markov property**, i.e., with

$$A_{t_0} = \sigma(\{X_t\}_{t \leq t_0})$$

for every $B \in \mathbb{B}_I$ and every $t_1 \geq t_0$,

$$P(X_{t_1} \in B | A_{t_0}) = P(X_{t_1} \in B | X_{t_0})$$

Definition 1.1.10. A Markov chain $(X_n)_n$ is homogeneous/stationary if the conditional probability $P(X_{n+1} = j | X_n = i)$ is the same for every n; in this case we denote this probability by p_{ij}.

Definition 1.1.11. Let $(X_n)_n$ be a given discrete-time homogeneous Markov chain. Then the matrix

$$\mathrm{P} = (p_{ij})_{i,j \in I}$$

with

$$p_{ij} = P(X_{n+1} = j | X_n = i)$$

is the matrix of one-step transition probabilities.

Remark 1.1.12. It may happen that $|I|$ is countably infinite; in this case we still use the term matrix.

The distribution of X_0 is called the initial distribution. Let $\mu = (\mu_i)_{i \in I}$ denote the distribution of X_0. Due to the Markov property,

$$P(X_0 = i_0, X_1 = i_1, ..., X_n = i_n)$$

$$= P(X_n = i_n | X_0 = i_0, X_1 = i_1, ..., X_{n-1} = i_{n-1}) \cdot P(X_0 = i_1, X_1 = i_1, ..., X_{n-1} = i_{n-1})$$

$$= P(X_n = i_n | X_0 = i_0, X_1 = i_1, ..., X_{n-1} = i_{n-1}) \cdot$$

$$\cdot P(X_{n-1} = i_{n-1} | X_0 = i_0, X_1 = i_1, ..., X_{n-2} = i_{n-2}) \cdot ... \cdot P(X_1 = i_1 | X_0 = i_0) \cdot P(X_0 = i_0)$$

$$= p_{i_{n-1} i_n} \cdot ... \cdot p_{i_0 i_1} \cdot \mu_{i_0}$$

Thus the initial distribution and the transition matrix together determine the Markov chain.

Let $m \in \mathbb{N}^+$ be arbitrary, and denote by $p_{ij}^{(n)} = P(X_{n+m} = j | X_m = i)$ the n-step transition probabilities (these are independent of m due to homogeneity).

Proposition 1.1.13. *(Chapman–Kolmogorov equation):*

$$p_{ij}^{(n+m)} = \sum_{k \in I} p_{ik}^{(n)} \cdot p_{kj}^{(m)}$$

Proof. Since $P(X_{n+m} = j, X_n = k) = P(X_n = k) \cdot P(X_{n+m} = j | X_n = k)$,

$$P(X_{n+m} = j | X_0 = i) = \sum_{k \in I} P(X_{n+m} = j, X_n = k | X_0 = i)$$

$$= \sum_{k \in I} P(X_n = k | X_0 = i) \cdot P(X_{n+m} = j | X_n = k, X_0 = i) = \sum_{k \in I} p_{ik}^{(n)} \cdot p_{kj}^{(m)}$$

□

Corollary 1.1.14. *$P^n = (p_{ij}^{(n)})_{i,j \in I}$, i.e., the matrix of n-step transition probabilities is exactly P^n.*

1.2 Hidden Markov Model

Definition 1.2.1. Let $X_n = X(n, \omega_X)$ be a discrete-time Markov chain, and $Y_n = Y(n, \omega_Y)$ a discrete-time stochastic process. Then the sequence (X_n, Y_n) is a (hidden) **Markov model** if for every n, for every $x_1, ..., x_n$, and for every Borel set A,

$$P(Y_n \in A | X_1 = x_1, ..., X_n = x_n) = P(Y_n \in A | X_n = x_n)$$

Definition 1.2.2. Let $X_t = X(t, \omega_X)$ be a continuous-time Markov chain, and $Y_t = Y(t, \omega_Y)$ a continuous-time stochastic process. Then $(X_t, Y_t)_{t \geq 0}$ is a (hidden) **Markov model** if for every $t_0 \in [0, \infty)$ and every Borel set $A \in B(I_Y)$,

$$P(Y_{t_0} \in A | \sigma(\{X_t : t \leq t_0\})) = P(Y_{t_0} \in A | \sigma(X_{t_0}))$$

In practice, the behavior of X_n is not directly observable (hence "hidden"), so we need to estimate it. In the discrete case, it is useful to represent the problem as a directed graph, where the two classes of nodes are the X_n's and the Y_n's, with an edge from every X_n to every X_m weighted by $P(X_n \to X_m)$, and an edge to every Y_k weighted by $P(Y_k | X_n)$ (for arbitrary $n, m, k \in \mathbb{N}^+$), together with a Start node from which edges lead into the X_n's according to the initial distribution.

The method presented here for estimating the X_n's requires knowledge of the type of the distributions involved.

Definition 1.2.3 (Posterior probability)**.** Let X be a hidden (not directly observable) state, and Y an observation. Then the conditional probability

$$P(X = x \mid Y = y)$$

is called the **posterior probability**. The posterior probability expresses how likely the given hidden state is, given the observation.

Using Bayes' theorem, the posterior probability can be computed as:

$$P(X = x \mid Y = y) = \frac{P(Y = y \mid X = x) \cdot P(X = x)}{P(Y = y)}$$

where:

- $P(X = x)$ is the **prior** probability,
- $P(Y = y \mid X = x)$ is the **likelihood**,
- $P(Y = y)$ is the **normalizing constant**.

1.3 Examples

You can find the following and similar examples in the book [WG04]

The Gambler's Ruin

Suppose in the beginning we have 2 dollars; every round we bet 1 dollar. With probability p, we win the game, otherwise we lose. The goal is to increase the capital to 4 dollars; as soon as this is reached, the game is over. Obviously, the game is also over if the capital is reduced to 0 dollars.

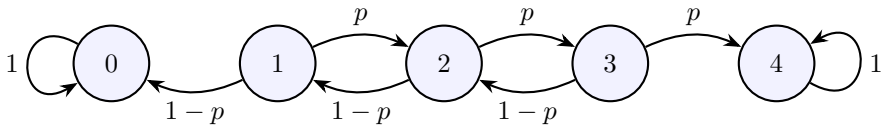
We can rephrase the above as a mathematical problem using the terminology of Markov chains: if we define X_t as the capital position after the tth game is played, then $X_0, X_1, \ldots$ may be viewed as a discrete-time stochastic process where $X_0 = 2$, $X_1 = 1$ with probability $1 - p$, and $X_1 = 3$ with probability p, etc. If there exists a t such that $X_t = 4$, then for every $t' \geq t$: $X_{t'} = 4$. Similarly, if there exists a t such that $X_t = 0$, then for every $t' \geq t$: $X_{t'} = 0$.

The amount of money after $t + 1$ plays depends only on the amount of money after t plays; therefore, we definitely have a Markov chain.

Since the rules do not change over time, this Markov chain is homogeneous. The transition matrix is given by:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1-p & 0 & p & 0 & 0 \\ 0 & 1-p & 0 & p & 0 \\ 0 & 0 & 1-p & 0 & p \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The corresponding state transition diagram of the Markov chain is shown below:



Solution

Question: Starting from $X_0 = 2$, what is the probability that the gambler reaches a capital of 4 before going bankrupt (reaching 0)?

Solution: States 0 and 4 are absorbing (once reached, the chain stays there forever), while states $1, 2, 3$ are transient. Let

$$h_i := P(\text{the chain reaches state 4 before state 0} \mid X_0 = i), \qquad i = 0, 1, 2, 3, 4.$$

By definition, $h_0 = 0$ and $h_4 = 1$.

For the transient states, we condition on the outcome of the first bet (first-step analysis). Since from state i the chain moves to $i + 1$ with probability p and to $i - 1$ with probability $1 - p$, and the future evolution after that step depends only on the new state (Markov property):

$$h_i = p \cdot h_{i+1} + (1 - p) \cdot h_{i-1}, \qquad i = 1, 2, 3.$$

Proposition 1.3.1. *For $p \neq \frac{1}{2}$,*

$$h_i = \frac{1 - \left(\frac{1-p}{p}\right)^i}{1 - \left(\frac{1-p}{p}\right)^4}, \qquad i = 0, 1, 2, 3, 4,$$

and for $p = \frac{1}{2}$,

$$h_i = \frac{i}{4}.$$

Proof. Rewrite the recursion as a homogeneous linear difference equation:

$$p \cdot h_{i+1} - h_i + (1 - p) \cdot h_{i-1} = 0.$$

Its characteristic equation is

$$px^2 - x + (1 - p) = 0,$$

whose roots are $x = 1$ and $x = \dfrac{1 - p}{p}$ (this follows since $x = 1$ is always a root, and the product of the roots is $\frac{1-p}{p}$ by Vieta's formulas).

Case $p \neq \frac{1}{2}$. The two roots are distinct, so the general solution of the recursion is

$$h_i = A + B \left(\frac{1 - p}{p}\right)^i$$

for some constants A, B. Using the boundary conditions:

$$h_0 = A + B = 0 \quad \Rightarrow \quad A = -B,$$

$$h_4 = A + B \left(\frac{1 - p}{p}\right)^4 = 1.$$

Substituting $A = -B$ into the second equation:

$$B \left(\left(\frac{1 - p}{p}\right)^4 - 1 \right) = 1 \quad \Rightarrow \quad B = \frac{1}{\left(\frac{1-p}{p}\right)^4 - 1}, \qquad A = -B.$$

Hence

$$h_i = A + B \left(\frac{1 - p}{p}\right)^i = B \left(\left(\frac{1 - p}{p}\right)^i - 1 \right) = \frac{\left(\frac{1-p}{p}\right)^i - 1}{\left(\frac{1-p}{p}\right)^4 - 1} = \frac{1 - \left(\frac{1-p}{p}\right)^i}{1 - \left(\frac{1-p}{p}\right)^4},$$

where the last equality follows by multiplying numerator and denominator by -1.

Case $p = \frac{1}{2}$. Here $\frac{1-p}{p} = 1$, so the characteristic equation has a double root at $x = 1$, and the general solution is $h_i = A + B \cdot i$. The boundary conditions $h_0 = 0$ and $h_4 = 1$ give $A = 0$ and $B = \frac{1}{4}$, so $h_i = \frac{i}{4}$. □

Applying the proposition with $i = 2$ (our starting capital) and $N = 4$:

$$h_2 = \begin{cases} \dfrac{1 - \left(\frac{1-p}{p}\right)^2}{1 - \left(\frac{1-p}{p}\right)^4}, & p \neq \frac{1}{2}, \\ \dfrac{1}{2}, & p = \frac{1}{2}. \end{cases}$$

For $p \neq \frac{1}{2}$, the fraction simplifies further: writing $r = \frac{1-p}{p}$, we have $1 - r^4 = (1 - r^2)(1 + r^2)$, so

$$h_2 = \frac{1 - r^2}{(1 - r^2)(1 + r^2)} = \frac{1}{1 + r^2} = \frac{1}{1 + \left(\frac{1-p}{p}\right)^2} = \frac{p^2}{p^2 + (1-p)^2}.$$

In particular, for a fair coin ($p = \frac{1}{2}$), the probability of reaching \$4 before going bankrupt starting from \$2 is exactly $\frac{1}{2}$, as expected by symmetry. The probability of ruin (reaching 0 before 4) is $1 - h_2$ in every case.

Painted Balls

An urn contains two unpainted balls. In each step, we choose a ball uniformly at random and then flip a fair coin:

- If the chosen ball is unpainted, we check the coin toss: if heads, we paint it red; if tails, we paint it black.
- If the chosen ball is already painted, we change its color to the opposite color (red to black, or black to red), regardless of the coin toss outcome.

This process can be modeled as a discrete-time stochastic process by recording the state of the urn as a triple (u, r, b), where:

$$u = |\{\text{unpainted balls}\}|, \quad r = |\{\text{red balls}\}|, \quad b = |\{\text{black balls}\}|$$

Since $u + r + b = 2$, there are 6 possible states in the state space S:

$$S = \{(2,0,0),\ (1,1,0),\ (1,0,1),\ (0,2,0),\ (0,1,1),\ (0,0,2)\}$$

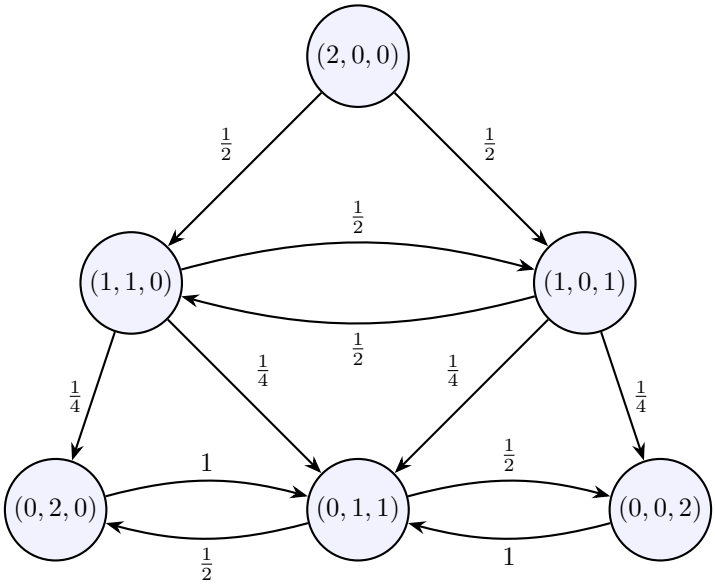
Notice that $X_0 = (2,0,0)$, and after the first step $X_1 = (1,1,0)$ or $X_1 = (1,0,1)$, each with probability $1/2$.

The state of the urn at time $t+1$ depends only on the state at time t, meaning the process satisfies the Markov property. Furthermore, since the rules do not change over time, this Markov chain is time-homogeneous.

Ordering the states as $S = ((2,0,0),(1,1,0),(1,0,1),(0,2,0),(0,1,1),(0,0,2))$, the transition matrix P is given by:

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

The corresponding state transition diagram is illustrated below:



Two Industries

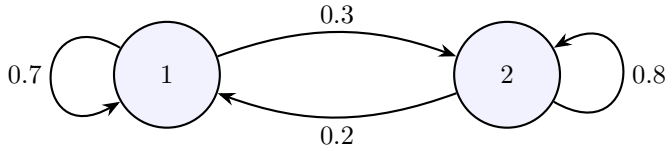
Suppose the world's entire food industry consists of two companies. If a customer last visited Company 1, there is a 70% chance they will visit it again next time. If someone visited Company 2, there is an 80% chance they will return there.

Question: If a customer is currently at Company 1, what is the probability that they will stay with Company 1 after 2 purchases from now? Under the same initial condition, what is the probability that they become a customer of Company 2 after 3 purchases from now?

Solution: A customer's choice can be modeled as a two-state time-homogeneous Markov chain, where the state space represents which company the customer visits. The transition matrix P is given by:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$$

The corresponding state transition diagram is shown below:



For the first question, we seek $P(X_2 = 1 \mid X_0 = 1)$. To find this 2-step transition probability from state 1 to state 1, we use the Chapman-Kolmogorov equations by computing P^2:

$$P^2 = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}^2 = \begin{pmatrix} 0.55 & 0.45 \\ 0.30 & 0.70 \end{pmatrix}$$

$$\Rightarrow P(X_2 = 1 \mid X_0 = 1) = P_{11}^2 = 0.55$$

For the second question, we seek $P(X_3 = 2 \mid X_0 = 1) = P_{12}^3$:

$$P^3 = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}^3 = \begin{pmatrix} 0.475 & 0.525 \\ 0.350 & 0.650 \end{pmatrix}$$

$$\Rightarrow P(X_3 = 2 \mid X_0 = 1) = P_{12}^3 = 0.525$$

Chapter 2

Steady-state distribution

In this chapter we will give an introduction to the ergodic theory in the shape of the steady-state distribution. If You imagine a random walk on a graph, the steady-state distribution says there is a given possibility for every vertex such that after a long time those are the process' actual position.

2.1 Concepts

Definition 2.1.1. Given states i, j, a path from i to j is a sequence of transitions begins in i and ends is j, such that each transition has a positive chance to happen. State i is reachable form state j if there is a path leading from i to j. Notation: $i \to j$.

i, j states are communicate if $i \to j$ and $j \to i$.

Notice that j is reachable from i in finite steps $\iff \exists n \geq 0$: $p_{ij}^{(n)} > 0$.

Definition 2.1.2. A set of states S in a Markov chain called closed if there is no state outside of S reachable from any state of S.

Definition 2.1.3. State i is absorbing if $p_{ii} = 1$ (in Gambler's ruin state 0 and 4 are absorbing). In other words, whenever we enter it we never leave it.

Definition 2.1.4. State i called transient if $\exists$ state j such that

$$i \to j \text{ and } j \nrightarrow i$$

In other words, there is a positive chance that whenever we leave state i we will not go there again, or the probability of ever returning to the state i is less than 1. Equivalently, the average time to return to then given state is finite.
A Markov chain called transient if every state of it transient.

Definition 2.1.5. State i called recurrent if the chance of getting back after reaching state i is 1. Equivalently, the expected number of returns is infinite (in the Urn example, for example, $(0, 2, 0)$ is recurrent). A Markov chain called recurrent if every state of it recurrent.

Definition 2.1.6. State i called periodic if

$$d(i) = gcd\{n \geq 0 : p_{ii}^{(n)} > 0\} > 1$$

State i called aperiodic if

$$d(i) = 1$$

A Markov chain called aperiodic if every state of it aperiodic (does not get trapped in loops).

Definition 2.1.7. A Markov chain called irreducible if every two states of it communicates.

Definition 2.1.8. A Markov chain called ergodic if irreducible, recurrent and aperiodic.

2.2 Discrete renewal equation

Relevant information for the following can be found in the book [Rén07]

Theorem 2.2.1. *(Discrete renewal equation) Let be given* $(a_k)_{k=0}^{\infty}, (u_k)_{k=0}^{\infty}, (b_k)_{k=0}^{\infty}$ *sequences such that*

$$b_k \geq 0, a_k \geq 0,$$

$$\sum a_k = 1, \sum(|k| \cdot a_k) < \infty, \sum(k \cdot a_k) > 0, \sum(|b_k|) < \infty,$$

$$gcd\{k|a_k > 0\} = 1$$

If the bounded sequence of real numbers $(u_n)_n$ *satisfy the renewal equations*

$$u_n - \sum(a_{n-k} \cdot u_k) = b_n$$

then $\lim u_n$ *exists and*

$$\lim u_n = \frac{\sum b_k}{\sum k \cdot a_k}$$

Proof. We can assume that $a_0 = 0, b_0 > 0$. Then the renewal equations:

$$u_n - \sum_{k=0}^{n} a_{n-k} \cdot u_k = b_n$$

can be written in the form

$$u_n - \sum_{k=0}^{n} a_k \cdot u_{n-k} = b_n$$

Using induction we have that $u_k \geq 0$. Indeed, it is true when $k = 0$. By the latest form of the equation, we have

$$(1 - a_0) \cdot u_n - (u_{n-1} - b_{n-1}) = b_n$$

By induction, we also have that $u_{n-1} - b_{n-1} > 0$ and $b_n \geq 0$, therefore $u_n \geq 0$. Let

$$\lambda = \limsup_{n \to \infty} u_n$$

We know by assumption that $\lambda < \infty$, then there exists $n_1 < n_2 < ...$ subsequence such that $\lim_{j \to \infty} u_{n_j} = \lambda$. Since we can assume that $a_1 > 0$, we have $\lim_{j \to \infty} u_{n_j - 1} = \lambda$. For this, let us consider the opposite case: suppose that $\lim \neq \lambda$. Then there exists $\lambda' < \lambda$ such that $u_{n_j - 1} < \lambda'$ for infinitely many j.

For the values

$$\epsilon = \frac{a_1 \cdot (\lambda - \lambda')}{4}$$

and

$$M = sup_n u_n$$

there exists N such that $\forall n \geq N$:

$$\sum_{k=0}^{\infty} a_k > 1 - \frac{\epsilon}{M}$$

There exists j such that

$$n_j \geq N, u_{n_j} > \lambda - \epsilon,$$
$$u_{n_j-1} < \lambda', 0 \leq b_{n_j} < \epsilon$$

and which for every $n \geq n_j - N$, we have

$$u_n < \lambda + \epsilon$$

Then $u_{n_j} \leq \sum_{k=0}^{n_j}(a_k \cdot u_{n_j-k} + \epsilon) < \sum_{k=0}^{N} a_k \cdot u_{n_j-k} + M \cdot \sum_{k=N+1}^{n_j}(a_k + \epsilon) < \sum_{k=0}^{N} a_k \cdot u_{n_j-k} + 2\epsilon <^{CS} (a_0 + ... + a_N) \cdot (\lambda + \epsilon) + a_1 \cdot \lambda' + 2\epsilon \leq (1 - a_1)(\lambda + \epsilon) + a_1 \cdot \lambda' + 2\epsilon < \lambda + 3\epsilon - a_1(\lambda - \lambda') = \lambda - \epsilon$, contradicting $u_{n_j} > \lambda - \epsilon$.

$$\Rightarrow \lim_{j\to\infty} u_{n_j-1} = \lambda$$

Similarly, $\forall d \in \mathbb{Z}^+$:

$$\lim_{j\to\infty} u_{n_j-d} = \lambda$$

Using the notation $r_n = a_{n+1} + a_{n+2} + ...$, we have

$$\sum_{k=0}^{\infty} k \cdot a_k = \sum_{n=0}^{\infty} r_n$$

and

$$a_1 = r_0 - r_1, a_2 = r_1 - r_2, ...$$

Hence $r_0 \cdot u_n + r_1 \cdot u_{n-1} + ... + r_n \cdot u_0 = r_0 \cdot u_{n-1} + r_1 \cdot u_{n-2} + ... + r_{n-1} \cdot u_0 + b_n$.

Let us denote the left by A_n, then we have $A_n = A_{n+1} + b_n$, where $A_0 = r_0 \cdot u_0 = (1 - a_0) \cdot u_0 = b_0$, therefore $A_n = \sum_{i=0}^{n} b_i$.

$\forall n : r_n \geq 0, u_n \geq 0 \Rightarrow$ for fixed $N, j > 0$:

$$r_0 \cdot u_{n_j} + r_1 \cdot u_{n_j-1} + ... + r_N \cdot u_{n_j-N} \leq A_n = \sum_{n=0}^{n_j} b_n$$

By taking the limit $j \to \infty$, we get

$$(r_0 + ... + r_N) \cdot \lambda \leq \sum_{n=0}^{\infty} b_n$$

By taking $N \to \infty$:

$$\lambda \leq \frac{\sum_{n=0}^{\infty} b_n}{\sum_{n=0}^{\infty} r_n}$$

Since $\sum_{n=0}^{\infty} r_n < \infty$, similarly for limsup, for $\mu = \liminf_{n\to\infty} u_n$, in the case $\lim_{j\to\infty} u_{n_j} = \mu$, we also have $\lim_{j\to\infty} u_{n_j-d} = \mu$ for every $d \in \mathbb{Z}^+$.

Let $g(N) = \sum_{n=N+1}^{\infty} r_n$, then $\lim_{N\to\infty} g(N) = 0$ and

$$\sum_{n=0}^{n_j} b_n \leq (r_0 \cdot u_{n_j} + ... + r_N \cdot u_{n_j-N}) + g(N) \cdot M$$

Taking $j \to \infty$:

$$\sum_{n=0}^{\infty} b_n \leq (r_0 + ... + r_N) \cdot \mu + g(N) \cdot M$$

Taking $N \to \infty$:

$$\sum_{n=0}^{\infty} b_n \leq \mu \cdot \sum_{n=0}^{\infty} r_n \iff \mu \geq \frac{\sum_{n=0}^{\infty} b_n}{\sum_{n=0}^{\infty} r_n} \geq \lambda$$

Hence $\lambda = \mu$

$$\Rightarrow \lim_{n\to\infty} u_n \text{ exists and } \lim_{n\to\infty} u_n = \frac{\sum_{n=0}^{\infty} b_n}{\sum_{n=0}^{\infty} r_n}$$

□

2.3 Ergodic theorem

Using the discrete renewal equation we can give a proof for the ergodic theorem of the Markov chains.

Theorem 2.3.1. *(Ergodic theorem) Let be given $(X_n)_n$ recurrent irreducible Markov chain,*

$$p_{ii}^n = \mathbb{P}(i \to i \text{ after } n \text{ steps})$$

$$f_{ii}^n = \mathbb{P}(i \to i \text{ for the first time } n)$$

Then

$$p_{ii}^n - \sum_{k=0}^{n} f_{ii}^{n-k} \cdot p_{ii}^k = \{1 \text{ if } n = 0,\ 0 \text{ otherwise}$$

and

$$\lim_{n\to\infty} p_{ii}^n = \frac{1}{\sum_{n=0}^{\infty} n \cdot f_{ii}^n}$$

As a consequence, we have

$$\pi_i = \lim_{n\to\infty} p_{ji}^n = \lim_{n\to\infty} p_{ii}^n$$

Proof. For the first part, use

$$p_{ji}^n = \sum_{v=0}^{n} f_{ji}^v \cdot p_{ii}^{n-v}$$

For the second part, apply the discrete renewal equation for the values

$$u_n = p_{ii}^n,\ a_n = f_{ii}^n \text{ and } b_n = \{\ 1 \text{ if } n = 0,\ 0 \text{ otherwise}$$

□

Remark 2.3.2. $(\pi_i)_i$ called the steady-state distribution.

2.4 Steady-state distribution

In this section we will see an application for the steady-state distribution.

Let us consider we have an ergodic stationary Markov chain with transition matrix P. As we seen previously, there exists a $(\pi_j)_j$ steady-state distribution for the process, that is the probability of being in state j after a long time

$$\lim_{n\to\infty} P_{ij}^n = \pi_j$$

Every convergent sequence is Cauchy, thereby $p_{ij}^{n+1} \approx p_{ij}^n \approx \pi_j$ as n grows. By Chapman-Kolmogorov,

$$\pi_j \approx p_{ij}^{n+1} = \sum_{k=1}^{s} p_{ik}^n \cdot p_{kj} \approx \sum_{k=1}^{s} \pi_k \cdot p_{kj}$$

$$\iff \pi = \pi \cdot P$$

It has infinite number of solutions (P is not fully ranked, $\mathbb{1}$ one of its eigenvectors). Note that for every n

$$p_{i1} + p_{i2} + ... + p_{is} = 1$$

Taking $n \to \infty$

$$\pi_1 + ... + \pi_s = 1$$

Using this condition allows us to get the solution we are really interested in.

Take the example of the two industries we talked about earlier, which defines a two-state ergodic Markov chain:

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$$

$\Rightarrow$

$$\begin{pmatrix} \pi_1 & \pi_2 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix} = \begin{pmatrix} \pi_1 & \pi_2 \end{pmatrix}$$

It gives us an equation system from we get $\pi_1 = 0.4$ and $\pi_2 = 0.6$ wich says that regardless someones current customer status, there is a 0.4 chance that after a long time they'll be a company1 customer, and a 0.6 chance being a company2 customer.

Chapter 3

Estimating the Hidden Markov Model

3.1 Expectation-Maximization Algorithm

In practice, due to the complicated form of the distribution, it is often difficult to give an ML estimate for the parameter of the distribution belonging to a hidden Markov model, so we need to approximate it. One possible approximation is given by the following algorithm:

Let a (discrete-time) hidden Markov model $(X_n, Y_n)_{(1\leq n\leq m)}$ be given, with observations $y = (y_1, ..., y_m)$ and corresponding hidden states $x = (x_1, ..., x_m)$, and distribution

$$(X, Y) \sim p((x, y)|\theta) = p_\theta(x, y) = \exp(\eta(\theta) \cdot s(x, y)) \cdot h(x, y)/c(\theta),$$

where η is a monotone function, $\Theta \subseteq \mathbb{R}^n$ is an open set representing the possible parameters of the distribution type under consideration,

$$s : \mathbb{R}^d \to \mathbb{R}^k$$

is a sufficient statistic,

$$h : \mathbb{R}^d \to \mathbb{R}^{\geq 0},$$
$$c : \Theta \to \mathbb{R}^{\geq 0}$$

is the normalizing function. That is,

$$c(\theta) = \int_{(X,Y)} (\exp(\eta(\theta) \cdot s(x, y)) \cdot h(x, y))\, d(x, y)$$

Our goal is to find $\theta_{MLE} \in \Theta$ such that

$$\theta_{MLE} \in \arg\max_\theta(p_\theta(y)),$$

i.e., it maximizes the likelihood function of the marginal distribution

$$p(\theta) = p_\theta(y) = \int p_\theta(x, y)dx$$

(or at least approximates a global maximum of it reasonably well). We may thus assume $\eta(\theta) = \theta$.

Proposition 3.1.1. $\nabla \log(c(\theta)) = \mathbb{E}_\theta s(x, y)$

Proof. For every j,

$$(\partial/\partial\theta_j)\log(c(\theta)) = \frac{1}{c(\theta)} \cdot (\partial/\partial\theta_j)c(\theta)$$

$$= \frac{1}{c(\theta)} \int s_j(x,y)p_\theta(x)d(x,y) = \mathbb{E}_\theta s_j(x,y)$$

(we assumed that differentiation and integration can be interchanged) □

Corollary 3.1.2. *if θ is an extremum of the likelihood function, then*

$$\mathbb{E}_\theta(s(X,Y)|Y=y) = \mathbb{E}_\theta(s(X,Y))$$

Proof.

$$0 = (\partial/\partial\theta_i)\log(p_\theta(y)) = \frac{1}{p(\theta)} \cdot \int (\partial/\partial\theta_i)p_\theta(x,y)dx$$

$$= \frac{1}{p(\theta)} \cdot \int (\partial/\partial\theta_i)\exp(\theta s(x,y) - \log(c(\theta))) \cdot h(x,y)dx$$

$$= \frac{1}{p(\theta)} \cdot \int (s_i(x,y) - (\partial/\partial\theta_i)\log(c(\theta))) \cdot p_\theta(x,y)$$

$$= \frac{1}{p(\theta)} \cdot \int (s_i(x,y) - \mathbb{E}_\theta s_i(X,Y)) \cdot p_\theta(x,y)$$

$$= \int s_i(x,y) \cdot p_\theta(x|y) - \mathbb{E}_\theta s_i(X,Y)$$

$$= \mathbb{E}_\theta(s_i(X,Y)|Y=y) - \mathbb{E}_\theta s_i(X,Y) \iff \mathbb{E}_\theta(s(X,Y)|Y=y) = \mathbb{E}_\theta(s(X,Y))$$

□

For every t, let

$$Q(\theta|\theta_t) = \mathbb{E}_{\theta_t}(\log(p_\theta(X,Y))|Y=y)$$

The idea of the algorithm is to start from an arbitrary estimate $\theta_0 \in \Theta$ and, at every iteration, improve it by solving the recursion

$$\theta_{t+1} = \arg\max_\theta Q(\theta|\theta_t)$$

Note that, by Proposition 3.1.1,

$$\nabla Q(\theta|\theta_t) = \nabla(\theta \cdot \mathbb{E}_{\theta_t}(s(X,Y)|Y=y) - \log(c(\theta)) - \text{constant})$$

$$= \mathbb{E}_{\theta_t}(s(X,Y)|Y=y) - \mathbb{E}_\theta(s(X,Y))$$

Thus, to solve the original recursion, we need to solve the recursion

$$\mathbb{E}_{\theta_{t-1}}(s(X,Y)|Y=y) = \mathbb{E}_{\theta_t}(s(X,Y))$$

Rule of thumb: we iterate until, for a fixed t, $\mathbb{E}_{\theta_t}(s(X,Y)|Y=y)$ and $\mathbb{E}_{\theta_t}(s(X,Y))$ are approximately equal (say, their difference is $< 10^{-5}$), since by the corollary above the estimate then approaches an extremum. Convergence follows from the fact that the sequence of values corresponding to the θ_t's is monotone and bounded; indeed, due to the recursion

$$\mathbb{E}_{\theta_{t-1}}(s(X,Y)|Y=y) = \mathbb{E}_{\theta_t}(s(X,Y))$$

and because the sequence

$$Q(\theta|\theta_t) = \mathbb{E}_{\theta_t}(\log(p_\theta(X,Y))|Y=y)$$

is convergent, it follows that there exists t such that $\mathbb{E}_{\theta_t}(s(X,Y)|Y=y)$ and $\mathbb{E}_{\theta_t}(s(X,Y))$ are approximately equal (because the sequences formed by these terms have the same limit).

The resulting extremum is not necessarily a global maximum; for this reason the algorithm should be run starting from several different initial values θ_0. We take the best estimate to be the presumed global maximum.

3.2 Bayes Filter

In the hidden Markov model appearing in the EM algorithm, we do not know the values

$$p_\theta(X, Y|Y = y),$$

so we would like to approximate them. One possible approximation is given by the Bayes filter:

Suppose the system of equations describing the hidden Markov model has the form

$$X_k = f(X_{k-1}, u_k, w_k)$$

$$Y_k = g(X_k, n_k)$$

where $u_k \in \mathbb{R}^N$ is the control input, and w_k and n_k are noise terms. Our goal is to give an estimate of the posterior probabilities

$$bel(X_k) = p(X_k|u_{1:k}, Y_{1:k})$$

for which we derive a recursion. Clearly $bel(X_0) = p(X_0)$.

Lemma 3.2.1.

$$p(x|y, z) = \frac{p(y|x, z) \cdot p(x|z)}{p(y|z)}$$

Proof.

$$\frac{p(y|x, z) \cdot p(x|z)}{p(y|z)} = \frac{p(y|x, z) \cdot p(x, z)}{p(z) \cdot p(y|z)} = \frac{p(y|z, x) \cdot p(z, x)}{p(y, z)} = \frac{p(x, y, z)}{p(y, z)} = p(x|y, z)$$

□

Proposition 3.2.2. *There exists a constant η such that*

$$bel(X_k) = \eta \cdot p(Y_k|X_k) \int p(X_k|X_{k-1}, u_k) \cdot bel(X_{k-1})\, dX_{k-1}$$

Proof. By the lemma above and the Markov property:

$$\begin{aligned}
bel(X_k) &= p(X_k|u_{1:k}, Y_{1:k}) \\
&= p(Y_k|X_k, u_{1:k}, Y_{1:k-1}) \cdot \frac{p(X_k|u_{1:k}, Y_{1:k-1})}{p(Y_k|u_{1:k}, Y_{1:k-1})} \\
&= \eta \cdot p(Y_k|X_k, u_{1:k}, Y_{1:k-1}) \cdot p(X_k|u_{1:k}, Y_{1:k-1}) \\
&= \eta \cdot p(Y_k|X_k) \cdot p(X_k|u_{1:k}, Y_{1:k-1}) \\
&= \eta \cdot p(Y_k|X_k) \cdot \int p(X_k, X_{k-1}|u_{1:k}, Y_{1:k-1})\, dX_{k-1} \\
&= \eta \cdot p(Y_k|X_k) \cdot \int p(X_k|X_{k-1}, u_{1:k}, Y_{1:k-1}) \cdot p(X_{k-1}|u_{1:k}, Y_{1:k-1})\, dX_{k-1} \\
&= \eta \cdot p(Y_k|X_k) \cdot \int p(X_k|X_{k-1}, u_k) \cdot p(X_{k-1}|u_{1:k-1}, Y_{1:k-1})\, dX_{k-1} \\
&= \eta \cdot p(Y_k|X_k) \int p(X_k|X_{k-1}, u_k) \cdot bel(X_{k-1})\, dX_{k-1}
\end{aligned}$$

□

Remark 3.2.3. Here $bel(X_k)$ denotes the posterior probability of state X_k at time k, given all observations $(Y_{1:k})$ and controls $(u_{1:k})$ up to that point.

Remark 3.2.4. **Application: Market making and estimating the true price.**

The Bayes filter provides a general framework for estimating the true value of an asset from noisy market data. A market maker's task is to set bid and ask prices. The true price v_t is not directly observable, but it can be inferred from the prices of trades.

Model:

$$v_t = v_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_v^2)$$
$$p_t = v_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_p^2)$$

The Bayes filter (in this case, the Kalman filter) gives an estimate of the true price ($\hat{v}_t$) and its uncertainty (σ_t). The market maker's strategy is then:

$$bid_t = \hat{v}_t - \delta - \gamma \cdot \sigma_t$$
$$ask_t = \hat{v}_t + \delta + \gamma \cdot \sigma_t$$

where δ covers transaction costs and γ is the risk premium. The larger the uncertainty (σ_t), the wider the bid-ask spread.

3.3 Convergence of the Bayes Filter

In this section we study the convergence of the Bayes filter. We show that the posterior distribution produced by the filter forgets the initial condition at an exponential rate, provided the system satisfies a so-called weak mixing condition. The proof rests on a contraction property of the filter operator in the projective metric.

Basic notions and notation

Let $(X_n)_{n\geq 0}$ be a Markov chain on a state space $\mathcal{X}$, and let $(Y_n)_{n\geq 1}$ be the sequence of observations. We assume the model has the following form:

- $X_0 \sim \mu_0$, the initial distribution,
- $(X_n \mid X_{n-1} = x) \sim Q(\cdot \mid x)$, the transition kernel,
- $(Y_n \mid X_n = x) \sim P(\cdot \mid x)$, the emission kernel.

The goal of the filter is to compute the posterior distribution, denoted by π_n:

$$\pi_n(f) := \mathbb{E}[f(X_n) \mid Y_1, \ldots, Y_n]$$

Let $\mathcal{P}(\mathcal{X})$ denote the space of probability measures on $\mathcal{X}$. The recursion of the filter consists of two steps.

Prediction (control update):

$$\pi_{n|n-1}(f) = \int_{\mathcal{X}} f(x)\, \pi_{n-1}(Q(\cdot \mid x))\, dx$$

Correction (measurement update):

$$\pi_n(f) = \frac{\pi_{n|n-1}(f \cdot p(Y_n \mid \cdot))}{\pi_{n|n-1}(p(Y_n \mid \cdot))}$$

where $p(y \mid x)$ is the emission density function.

The convergence theorem

Theorem 3.3.1 (Convergence of the Bayes filter)**.** *Suppose the following* ***weak mixing condition*** *holds:*

There exist $\alpha \in (0,1)$ *and* $C < \infty$ *such that, for the transition kernel* Q*, for every* $x, x' \in \mathcal{X}$ *and* n*,*

$$\rho(Q(\cdot \mid x), Q(\cdot \mid x')) \leq C \cdot \alpha^n,$$

where ρ *is the Hilbert projective metric.*

Then the Bayes filter operator is a contraction in the projective metric, that is, for the outputs of the filter started from any two different initial distributions (π_0 *and* π_0'*):*

$$\rho(\pi_n, \pi_n') \leq C \cdot \alpha^n \cdot \rho(\pi_0, \pi_0')$$

Consequently, in total variation norm:

$$\|\pi_n - \pi_n'\|_{TV} \longrightarrow 0 \quad (n \to \infty)$$

and the rate of convergence is exponential:

$$\|\pi_n - \pi_n'\|_{TV} = O(\alpha^n)$$

Tools for the proof

The projective metric

Definition 3.3.2 (Projective metric)**.** Let μ be an arbitrary σ-finite measure, and let $p, q \in L^1(\mu)$ be strictly positive density functions. The **Hilbert projective metric** is defined as follows:

$$\rho(p,q) := \log\left(\sup_A \frac{p(A)/\mu(A)}{q(A)/\mu(A)} \cdot \sup_A \frac{q(A)/\mu(A)}{p(A)/\mu(A)}\right)$$

or, equivalently:

$$\rho(p,q) = \sup_{x,y}\left|\log\frac{p(x)}{q(x)} - \log\frac{p(y)}{q(y)}\right|$$

Remark 3.3.3. The projective metric is invariant under multiplication by a constant: $\rho(\lambda p, \mu q) = \rho(p,q)$ for every $\lambda, \mu > 0$. Furthermore, its relationship to the total variation norm is:

$$\|p - q\|_{TV} \leq \frac{2}{\sqrt{e^{\rho(p,q)} + 1}} \cdot \rho(p,q)$$

from which it follows that convergence in the projective metric implies convergence in total variation.

Lemma 3.3.4 (Contraction of the correction operator)**.** *Let* $p, q \in \mathcal{P}(\mathcal{X})$ *be two density functions, and let* y *be a fixed observation. Define the correction operator:*

$$(T_y p)(x) := \frac{p(x) \cdot p(y \mid x)}{\int_{\mathcal{X}} p(z) \cdot p(y \mid z)\, dz}$$

Then, in the projective metric, the correction operator does not increase the distance, that is,

$$\rho(T_y p, T_y q) \leq \rho(p,q)$$

Proof. From the definition of the projective metric:

$$\rho(T_y p, T_y q) = \rho(p \cdot p(y \mid \cdot), q \cdot p(y \mid \cdot))$$

Since multiplication by a positive function is always a contraction in the projective metric (that is, $\rho(f \cdot p, f \cdot q) \leq \rho(p,q)$), we get:

$$\rho(T_y p, T_y q) \leq \rho(p,q)$$

This step therefore causes no loss of information in the forgetting of the initial condition. □

Lemma 3.3.5 (Contraction of the prediction operator)**.** *Let Q be the transition kernel. The prediction operator is:*

$$(Pp)(x) := \int_{\mathcal{X}} Q(x \mid z)\, p(z)\, dz$$

Then, under the weak mixing condition:

$$\rho(Pp, Pq) \leq C \cdot \alpha^n \cdot \rho(p, q)$$

Proof. The proof is a direct application of the mixing condition. By assumption:

$$\rho(Q(\cdot \mid x), Q(\cdot \mid x')) \leq C \cdot \alpha^n$$

Since Pp and Pq are mixtures of Q with respect to p and q, by convexity of the projective metric:

$$\rho(Pp, Pq) \leq \sup_{x,x'} \rho(Q(\cdot \mid x), Q(\cdot \mid x')) \leq C \cdot \alpha^n \cdot \rho(p, q)$$

The relationship between $\rho(Pp, Pq)$ and $\rho(p, q)$ is guaranteed by the mixing condition. □

Proof of the theorem

Proof. Let π_n and π'_n be the outputs of two filters, started from two different initial distributions (π_0 and π'_0).

The recursion of the filter is the composition of prediction and correction:

$$\pi_n = T_{Y_n} \circ P \circ T_{Y_{n-1}} \circ P \circ \cdots \circ T_{Y_1} \circ P(\pi_0)$$

By the lemmas above, every single step is a contraction:

1. The prediction operator P is a contraction: $\rho(Pp, Pq) \leq C \cdot \alpha^n \cdot \rho(p, q)$
2. The correction operator T_y does not increase the distance: $\rho(T_y p, T_y q) \leq \rho(p, q)$

Applying these recursively:

$$\rho(\pi_n, \pi'_n) \leq \rho(P\pi_{n-1}, P\pi'_{n-1}) \leq C \cdot \alpha^n \cdot \rho(\pi_0, \pi'_0)$$

The right-hand side tends to 0 as $n \to \infty$, since $\alpha \in (0, 1)$.

Since the projective metric and the total variation norm satisfy the relationship:

$$\|p - q\|_{TV} \leq \frac{2}{\sqrt{e^{\rho(p,q)} + 1}} \cdot \rho(p, q)$$

convergence in the projective metric implies convergence in total variation:

$$\|\pi_n - \pi'_n\|_{TV} \longrightarrow 0 \quad (n \to \infty)$$

The rate of convergence, following from the $C \cdot \alpha^n$ form of the right-hand side, is exponential:

$$\|\pi_n - \pi'_n\|_{TV} = O(\alpha^n)$$

This completes the proof of the theorem. □

Remark 3.3.6. The mixing condition guarantees that the Markov chain forgets its initial state. This condition holds in most practical applications (e.g. stationary, ergodic processes).

Remark 3.3.7. If the observations are unboundedly informative, the correction step can lock the filter into a given state, so that convergence can then only occur through the prediction step.

Chapter 4

Discrete-State Hidden Markov Model

4.1 Forward-Backward Algorithm

One of the most fundamental algorithms for discrete-time hidden Markov models is the Forward-Backward algorithm, which lets us compute, given an observation sequence, the posterior probabilities of the hidden states, as well as the quantities needed for the E-step of the EM algorithm.

Let a discrete-time hidden Markov model $(X_n, Y_n)_{n=1}^T$ be given, where $X_n \in \{1, 2, \ldots, N\}$ are the hidden states and $Y_n \in \{1, 2, \ldots, M\}$ are the observations. The parameters of the model are:

- $\pi_i = P(X_1 = i)$, the initial distribution,
- $A_{ij} = P(X_{n+1} = j | X_n = i)$, the matrix of transition probabilities,
- $B_{ik} = P(Y_n = k | X_n = i)$, the matrix of emission probabilities.

The parameters of the model are thus $\theta = (\pi, A, B)$. Our goal is to compute, given the observations $y_{1:T} = (y_1, \ldots, y_T)$, the following quantities:

1. The filtered probabilities: $\alpha_n(i) = P(X_n = i, y_{1:n})$
2. The smoothed probabilities: $\beta_n(i) = P(y_{n+1:T} | X_n = i)$
3. The posterior state probabilities: $bel_n(i) = P(X_n = i | y_{1:T})$
4. The posterior transition probabilities: $\xi_n(i, j) = P(X_n = i, X_{n+1} = j | y_{1:T})$

Here $bel_n(i)$ denotes the posterior probability of state i at time n, given the entire observation sequence.

Forward algorithm

The Forward algorithm computes the quantities $\alpha_n(i)$ recursively. The initialization follows from the law of total probability:

$$\alpha_1(i) = P(X_1 = i, Y_1 = y_1) = P(Y_1 = y_1 | X_1 = i) \cdot P(X_1 = i) = B_{iy_1} \pi_i.$$

For the recursion, we apply the Markov property:

$$\begin{aligned}
\alpha_{n+1}(j) &= P(X_{n+1} = j, y_{1:n+1}) \\
&= P(y_{n+1}|X_{n+1} = j, y_{1:n}) \cdot P(X_{n+1} = j, y_{1:n}) \\
&= B_{jy_{n+1}} \sum_{i=1}^{N} P(X_{n+1} = j|X_n = i) \cdot P(X_n = i, y_{1:n}) \\
&= B_{jy_{n+1}} \sum_{i=1}^{N} A_{ij}\alpha_n(i).
\end{aligned}$$

Thus the Forward algorithm is:

$$\begin{aligned}
\alpha_1(i) &= \pi_i \cdot B_{iy_1} \quad \text{(initialization)} \\
\alpha_{n+1}(j) &= B_{jy_{n+1}} \sum_{i=1}^{N} \alpha_n(i)A_{ij} \quad \text{(recursion)}
\end{aligned}$$

The total probability of the observations can be computed from the α variables:

$$P(y_{1:T}) = \sum_{i=1}^{N} \alpha_T(i)$$

Backward algorithm

The Backward algorithm computes the quantities $\beta_n(i)$ recursively. The initialization is trivial: $P(\emptyset|X_T = i) = 1$, so $\beta_T(i) = 1$. For the recursion:

$$\begin{aligned}
\beta_n(i) &= P(y_{n+1:T}|X_n = i) \\
&= \sum_{j=1}^{N} P(X_{n+1} = j, y_{n+1:T}|X_n = i) \\
&= \sum_{j=1}^{N} P(X_{n+1} = j|X_n = i) \cdot P(y_{n+1:T}|X_{n+1} = j) \\
&= \sum_{j=1}^{N} A_{ij} \cdot P(y_{n+1}|X_{n+1} = j) \cdot P(y_{n+2:T}|X_{n+1} = j) \\
&= \sum_{j=1}^{N} A_{ij}B_{jy_{n+1}}\beta_{n+1}(j).
\end{aligned}$$

Thus the Backward algorithm is:

$$\begin{aligned}
\beta_T(i) &= 1 \quad \text{(initialization)} \\
\beta_n(i) &= \sum_{j=1}^{N} A_{ij}B_{jy_{n+1}}\beta_{n+1}(j) \quad \text{(recursion)}
\end{aligned}$$

Posterior state and transition probabilities

Using the α and β variables, we can compute the posterior state probabilities $bel_n(i)$. By Bayes' theorem and conditional independence:

$$\begin{aligned}
P(X_n = i|y_{1:T}) &= \frac{P(X_n = i, y_{1:T})}{P(y_{1:T})} \\
&= \frac{P(X_n = i, y_{1:n}) \cdot P(y_{n+1:T}|X_n = i)}{P(y_{1:T})} \\
&= \frac{\alpha_n(i)\beta_n(i)}{P(y_{1:T})}.
\end{aligned}$$

That is:

$$bel_n(i) = P(X_n = i|y_{1:T}) = \frac{\alpha_n(i)\beta_n(i)}{\sum_{j=1}^{N} \alpha_T(j)}$$

For the posterior transition probabilities, applying Bayes' theorem and the Markov property:

$$\begin{aligned} P(X_n = i, X_{n+1} = j, y_{1:T}) &= P(X_n = i, y_{1:n}) \cdot P(X_{n+1} = j|X_n = i) \\ &\quad \cdot P(y_{n+1}|X_{n+1} = j) \cdot P(y_{n+2:T}|X_{n+1} = j) \\ &= \alpha_n(i) \cdot A_{ij} \cdot B_{jy_{n+1}} \cdot \beta_{n+1}(j). \end{aligned}$$

Dividing by $P(y_{1:T})$:

$$\xi_n(i,j) = P(X_n = i, X_{n+1} = j|y_{1:T}) = \frac{\alpha_n(i)A_{ij}B_{jy_{n+1}}\beta_{n+1}(j)}{\sum_{k=1}^{N} \alpha_T(k)}$$

Scaled Forward-Backward algorithm

Let $\alpha_n(i) = P(X_n = i, y_{1:n})$ and $\beta_n(i) = P(y_{n+1:T} \mid X_n = i)$. The scaled variables are:

$$\hat{\alpha}_n(i) = \frac{\alpha_n(i)}{\sum_{j=1}^{N} \alpha_n(j)}, \quad \hat{\beta}_n(i) = \frac{\beta_n(i)}{\sum_{j=1}^{N} \beta_n(j)}$$

The scaling factors are:

$$c_n = \sum_{j=1}^{N} \alpha_n(j)$$

Then the log-likelihood is:

$$\log P(y_{1:T}) = \sum_{n=1}^{T} \log c_n$$

Convergence of the Forward-Backward algorithm

Theorem 4.1.1 (Forward-Backward convergence)**.** *Suppose the Markov chain is irreducible and aperiodic, and the emission densities are positive. Then the posterior probabilities computed by the scaled Forward-Backward algorithm,*

$$\pi_n(i) = \hat{\alpha}_n(i)\hat{\beta}_n(i)$$

converge to the true posterior probabilities, that is:

$$\pi_n(i) \longrightarrow P(X_n = i \mid y_{1:T}) \quad (T \to \infty)$$

The rate of convergence is exponential:

$$|\pi_n(i) - P(X_n = i \mid y_{1:T})| = O(\alpha^n)$$

for some $\alpha \in (0, 1)$.

Proof. **1. Convergence of the filter:** The Bayes filter operator is a contraction in total variation norm, provided the mixing condition holds. This means the predictive distribution converges to the stationary distribution:

$$\|\pi_{n|n-1} - \pi^*\|_{TV} \leq C \cdot \alpha^n$$

2. Convergence of the smoothing step: The variables $\beta_n(i)$ give the probability of future observations. By the mixing condition:

$$|\beta_n(i) - \beta_n(j)| \leq C \cdot \alpha^{T-n}$$

3. Convergence of the posterior probabilities: From Bayes' theorem:

$$\pi_n(i) = \frac{\alpha_n(i)\beta_n(i)}{\sum_j \alpha_n(j)\beta_n(j)}$$

Substituting the bounds above:

$$|\pi_n(i) - P(X_n = i \mid y_{1:T})| \leq C \cdot \alpha^{\min(n, T-n)}$$

Since $\alpha \in (0, 1)$, the right-hand side tends to 0 as $T \to \infty$. □

4.2 Baum-Welch Algorithm

The Baum-Welch algorithm is the special case of the EM algorithm for parameter estimation in hidden Markov models. The intuitive meaning of the re-estimation formulas is:

- $\pi_i^{\text{new}} = bel_1(i)$: the expected probability that state i is the initial state,
- $A_{ij}^{\text{new}} = \dfrac{\sum_{n=1}^{T-1} \xi_n(i,j)}{\sum_{n=1}^{T-1} bel_n(i)}$: the expected number of transitions from i to j, divided by the expected number of transitions out of i,
- $B_{ik}^{\text{new}} = \dfrac{\sum_{n=1}^{T} bel_n(i) \cdot I(y_n = k)}{\sum_{n=1}^{T} bel_n(i)}$: the expected number of times observation k occurs while in state i, divided by the expected time spent in state i.

Formally:

$$\begin{aligned}
\pi_i^{\text{new}} &= bel_1(i) \\
A_{ij}^{\text{new}} &= \frac{\sum_{n=1}^{T-1} \xi_n(i,j)}{\sum_{n=1}^{T-1} bel_n(i)} \\
B_{ik}^{\text{new}} &= \frac{\sum_{n=1}^{T} bel_n(i) \cdot I(y_n = k)}{\sum_{n=1}^{T} bel_n(i)}
\end{aligned}$$

The Baum-Welch algorithm monotonically increases the log-likelihood of the observations:

$$P(y_{1:T}|\theta^{\text{new}}) \geq P(y_{1:T}|\theta)$$

and converges to a local maximum. The proof rests on the general convergence theorem for the EM algorithm.

Monotone increase of the likelihood

Theorem 4.2.1 (Baum-Welch convergence)**.** *Let Λ be the old model and Λ' the new model. The Baum-Welch algorithm increases (or at least does not decrease) the log-likelihood of the observations at every iteration:*

$$\log P(y_{1:T} \mid \Lambda') \geq \log P(y_{1:T} \mid \Lambda)$$

Equality holds if and only if $\Lambda = \Lambda'$.

Proof. The proof rests on the non-negativity of the Kullback-Leibler divergence.

1. Expanding the likelihood ratio: We expand the log-likelihood ratio using the quantity $P(X_{1:T} \mid y_{1:T}, \Lambda')$:

$$\log \frac{P(y_{1:T} \mid \Lambda)}{P(y_{1:T} \mid \Lambda')} = \sum_{X_{1:T}} P(X_{1:T} \mid y_{1:T}, \Lambda') \log \frac{P(y_{1:T}, X_{1:T} \mid \Lambda)}{P(y_{1:T}, X_{1:T} \mid \Lambda')} \cdot \frac{P(X_{1:T} \mid y_{1:T}, \Lambda')}{P(X_{1:T} \mid y_{1:T}, \Lambda)}$$

2. Splitting into two terms: The expression above splits into two terms:

- $L(\Lambda, \Lambda') = \sum_{X_{1:T}} P(X_{1:T} \mid y_{1:T}, \Lambda') \log \frac{P(y_{1:T}, X_{1:T}|\Lambda)}{P(y_{1:T}, X_{1:T}|\Lambda')}$
- $KL(P(X_{1:T} \mid y_{1:T}, \Lambda') \| P(X_{1:T} \mid y_{1:T}, \Lambda)) \geq 0$

3. The Baum-Welch algorithm maximizes L: The M-step of the Baum-Welch algorithm maximizes exactly $L(\Lambda, \Lambda')$ with respect to Λ'. Therefore:

$$L(\Lambda', \Lambda') \geq L(\Lambda, \Lambda')$$

4. Conclusion:

$$\log \frac{P(y_{1:T} \mid \Lambda)}{P(y_{1:T} \mid \Lambda')} = L(\Lambda, \Lambda') + KL(\cdot) \geq 0$$

From which:

$$\log P(y_{1:T} \mid \Lambda') \geq \log P(y_{1:T} \mid \Lambda)$$

The likelihood therefore increases or stays the same at every iteration, until convergence is reached. □

Practical implementation and numerical stability

The full implementation of the Forward-Backward algorithm consists of the following steps:

1. **Forward step**: Compute $\alpha_1(i)$ for every i, then recursively compute $\alpha_n(i)$ for $n = 2, \ldots, T$.
2. **Backward step**: Compute $\beta_T(i)$ for every i, then recursively compute $\beta_n(i)$ for $n = T-1, \ldots, 1$.
3. **Computing the posteriors**: Compute $bel_n(i)$ and $\xi_n(i, j)$ for every n, i, j.
4. **Re-estimation**: Update the parameters using the Baum-Welch formulas.

The computational complexity of the algorithm is $O(T \cdot N^2)$, where T is the number of time steps and N is the number of states.

Remark 4.2.2. **Application: identifying market regimes.** Suppose we model the daily returns of a stock index (e.g. the S&P 500) using two hidden states:

- $X_t = 1$: calm market (low volatility, positive expected return)
- $X_t = 2$: volatile market (high volatility, negative or zero expected return)

The observations Y_t are the daily returns, which we model with a conditional normal distribution:

$$Y_t | X_t = i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad i = 1, 2$$

Using the Baum-Welch algorithm, we can estimate the model parameters $(\pi, A, \mu_1, \sigma_1, \mu_2, \sigma_2)$ from historical data. The Forward-Backward algorithm computes, for every time point, the posterior probabilities $bel_t(i) = P(X_t = i | y_{1:T})$, i.e. the probability that a given regime was present on that day.

Chapter 5

Continuous-State Hidden Markov Model

5.1 Kalman Filter

The Bayes filter algorithm can only be computed in closed form in special cases in practice. The two main limiting factors are:

- Carrying out the integral in the prediction step,
- Representing the posterior density function.

The Kalman filter is the closed-form solution of the Bayes filter for linear Gaussian systems. It is one of the most widely used filters in practice, thanks to its simplicity and optimality properties.

Linear Gaussian systems: The Kalman filter relies on the following three assumptions:

1. The state transition is linear with additive Gaussian noise:

$$x_t = A_t x_{t-1} + B_t u_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, R_t)$$

2. The observation model is linear with additive Gaussian noise:

$$z_t = C_t x_t + \delta_t, \quad \delta_t \sim \mathcal{N}(0, Q_t)$$

3. The initial state is Gaussian-distributed:

$$bel(x_0) = \mathcal{N}(x_0; \mu_0, \Sigma_0)$$

These conditions guarantee that the posterior distribution remains Gaussian at every time step.

The Kalman filter algorithm:

The filter represents the distribution $bel(x_t)$ by the mean μ_t and the covariance matrix Σ_t.

Prediction step:

$$\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$$
$$\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t$$

Correction step:

$$
\begin{aligned}
K_t &= \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \\
\mu_t &= \bar{\mu}_t + K_t(z_t - C_t \bar{\mu}_t) \\
\Sigma_t &= (I - K_t C_t)\bar{\Sigma}_t
\end{aligned}
$$

where K_t is the Kalman gain.

Derivation of the Kalman filter:

The prediction step follows from the prediction formula of the Bayes filter. Since both factors are Gaussian, their convolution is also Gaussian:

$$
\overline{bel}(x_t) = \int \mathcal{N}(x_t; A_t x_{t-1} + B_t u_t, R_t) \cdot \mathcal{N}(x_{t-1}; \mu_{t-1}, \Sigma_{t-1})\, dx_{t-1}
$$

The product is Gaussian, with parameters given by the prediction formulas above.

The correction step is an application of Bayes' theorem:

$$
bel(x_t) = \eta \cdot p(z_t|x_t) \cdot \overline{bel}(x_t) = \eta \cdot \mathcal{N}(z_t; C_t x_t, Q_t) \cdot \mathcal{N}(x_t; \bar{\mu}_t, \bar{\Sigma}_t)
$$

The product of two Gaussian distributions is also Gaussian, with parameters given by the correction formulas above.

Properties

- **Optimality:** The Kalman filter gives the estimate that is optimal in the minimum mean-squared-error sense, for linear Gaussian systems.
- **Recursivity:** The filter uses only the previous state estimate and the current observation, so it can be applied online.
- **Complexity:** Each step takes $O(n^3)$ time due to the matrix inversion, where n is the dimension of the state space.

Remark 5.1.1. **Applications of the Kalman filter:**

Trend following: Let p_t be the price of an asset. Suppose the price fluctuates around a hidden trend:

$$
\begin{aligned}
\mu_t &= \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2) \\
p_t &= \mu_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)
\end{aligned}
$$

In Kalman filter form: $x_t = \mu_t$, $A_t = 1$, $R_t = \sigma_\eta^2$, $C_t = 1$, $Q_t = \sigma_\varepsilon^2$. At every time step, the filter gives an estimate of the trend (μ_t) and its uncertainty (σ_t^2).

Volatility estimation: In stochastic volatility models, the log-volatility is the hidden state:

$$
\begin{aligned}
r_t &= \sigma_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1) \\
\log \sigma_t^2 &= \log \sigma_{t-1}^2 + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2)
\end{aligned}
$$

This model is nonlinear, so an EKF or UKF is needed for estimation. The estimated volatility can be used for option pricing and risk management.

Factor timing: In dynamic factor models, the factor loadings change over time:

$$
R_t = \alpha_t + \sum_{k=1}^{K} \beta_t^k F_t^k + \varepsilon_t, \quad \theta_t = \theta_{t-1} + \eta_t
$$

where $\theta_t = (\alpha_t, \beta_t^1, \ldots, \beta_t^K)^T$. The Kalman filter gives a dynamic estimate of the factor loadings, which allows for timing factor strategies.

Chapter 6

Quantum Markov chains

6.1 Concepts

Before our first definition, we introduce a basic notation used in quantum mechanics: $\langle\psi|$ denotes the representations of ψ as a row vector, $|\psi\rangle$ as a column vector. Using $\langle\psi|$ and $|\psi\rangle$ the same time means the inner or the outer product of two vectors, for example $\langle\psi|\,|\phi\rangle$ is the inner product of the vectors ψ, ϕ, and $|\psi\rangle\,\langle\phi|$ is the outer product.

† denotes the transposed conjugate.

Definition 6.1.1. An operator ρ acting on a Hilbert space called density operator if satisfies the following conditions:

- Hermitian: $\rho^\dagger = \rho$ (all observable probabilities are real numbers)
- Positive semi-definite: for any state $|\phi\rangle$, $\langle\phi|\,\rho\,|\phi\rangle \geq 0$ (the probabilities are non-negative)
- Unit trace: $\mathrm{Tr}(\rho) = 1$ (the probability of finding the system in any state sums to 1)

If a system is in a pure state $|\psi\rangle$ with probability 1, the density operator is the outer product $\rho = |\psi\rangle\,\langle\psi|$.

If a system is in a mixed state (has a probability $p_1 > 0$ being in state $|\psi_1\rangle$, a probability $p_2 > 0$ being in state $|\psi_2\rangle$, so on), the density operator is the weighted sum

$$\rho = \sum_i p_i \cdot |\psi_i\rangle\,\langle\psi_i|$$

Let our system be a complex n-dimensional vector space $H \sim \mathbb{C}^n$.

The expectation value defined as a functional, let us say $f : H \times H \to \mathbb{C}$. For be a valid expectation value, it must satisfy the linearity, positivity and normalization:

- $f(\alpha \cdot A + \beta \cdot B) = \alpha \cdot f(A) + \beta \cdot f(B)$ for every $\alpha, \beta \in \mathbb{C}$
- If A is positive semi-definite then $f(A) \geq 0$
- If $A = I$ then $f(A) = 1$

Definition 6.1.2. The expected value of a measurable observable A is given by

$$\mathbb{E}[A] = \mathrm{Tr}(A\rho)$$

The motivation for this definition given as follows:

The set of all $n \times n$ matrices forms a Hilbert space under the Hilbert-Schmidt product defined as

$$\langle X, Y\rangle = \mathrm{Tr}(X^\dagger Y)$$

By the matrix form of Riesz representation theorem, for every continuous linear functional $f(A)$ there exists a unique matrix R such that

$$f(A) = \mathrm{Tr}(R^\dagger A)$$

Since $f(A)$ is Hermitian, R is also Hermitian. R must be positive semi-definite. $f(I) = \mathrm{Tr}(R^\dagger I) = \mathrm{Tr}(R) = 1$, hence R have trace 1.

$$\Rightarrow R = \rho$$

Definition 6.1.3. Let H be a complex Hilbert space, denote by $B(H)$ the algebra of bounded linear operators on H. A unital $*$-subalgebra $M \subseteq B(H)$ is a von Neumann algebra if it equals with its own double commutant:

$$M = M''$$

where $M' = \{T \in B(H) \mid TX = XT \text{ for every } T \in M\}$.
Equivalently, M is closed in the weak operator topology.

Definition 6.1.4. Let M and N be von Neumann algebras acting on Hilbert spaces H and K, respectively. Let M_* and N_* denote their preduals (the space of normal linear functionals).
A quantum channel is a linear map

$$\Phi : M_* \to N_*$$

satisfying:

- Trace-preserving (Normalized): Φ maps states (positive normalized normal functions) to states. Equivalently, fot the identity 1_N, the dual map satisfies $\Phi^*(1_N) = 1_M$
- Completely positive: For every n, the amplified map

  $$\Phi \otimes id_n : M_* \otimes M_n(\mathbb{C}) \to N_* \otimes M_n(\mathbb{C})$$

  is positive.
- Normal: Φ is continuous with respect to the σ-weak topologies on the preduals. This guarantees a well-defined dual map $\Phi^* : N \to M$ which is normal, completely positive and unital.

Definition 6.1.5. A quantum Markov chain is a pair (E, ρ) with ρ a density matrix and E a quantum channel such that

$$E : B \otimes B \to B$$

is a completely positive trace-preserving map, and B a von Neumann algebra of bounded operators. It must obey the quantum Markov condition:

$$\mathrm{Tr}\, \rho(b_1 \otimes b_2) = \mathrm{Tr}\, \rho E(b_1, b_2)$$

for all $b_1, b_2 \in B$.

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